

The Asymptotic-Safety Third Corner in MTT: A Conditional Diagnostic Between Coherent Geometry and the String Branch

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Version 2

Abstract

Modal Triplet Theory currently approaches ultraviolet gravity through two partly developed routes: selected q79 heterotic geometry and projection-filtered four-dimensional constructions. Functional renormalization group asymptotic safety can serve as a third diagnostic corner, but it is not yet equivalent to either route. This paper gives a theorem-light map of the three descriptions. The q79 branch has exact finite algebraic data and a partial heterotic worldsheet contract; the physical visible-hidden bundle, continuum HYM operator, and infrared conformal field theory remain open. The FRG corner has conditional endpoint, conjugacy, truncation-error, and unstable-subspace theorems, but no selected MTT-to-coupling chart. Agreement of fixed points or critical exponents can therefore be used only after overlap domains, maps, conventions, and defects are supplied. The useful claim is a conditional triple diagnostic: independent encodings may test one selected upper construction. The paper does not claim equivalence of ultraviolet completions.

Version 2 Revision Note

Supersedes	Version 1.0, <i>A Third-Corner Shadow Bridge: Asymptotic Safety, the String Corner, and the Coherent Spine in Modal Triplet Theory</i> .
Reason	The earlier paper promoted assumed conjugacies into a triple-corner equivalence and used an invalid additive-error fixed-point lemma.
Resolution	All technical fixed-point claims are assigned to their owner papers. This paper now describes the three corners, their actual current status, and the maps and error certificates needed for comparison.
Retained result	A three-corner comparison is a useful consistency test when the encodings arise from one selected upper object and their overlap maps commute.
Remaining boundary	The physical q79 worldsheet contract, continuum geometry-to-operator map, and MTT-to-FRG coupling chart are all incomplete. No triple equivalence is claimed.

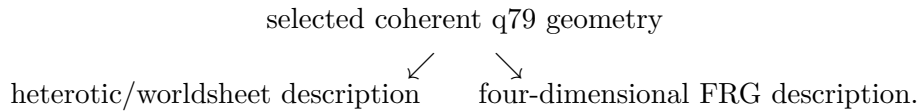
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1 The idea that survives

The motivating idea is good: one underlying construction can have several effective descriptions, and agreement among them is stronger evidence than success in one representation. In MTT the proposed corners are:



The correction concerns the arrows. They are not supplied by drawing the diagram. Each arrow must map states, operators, normalizations, and dynamics on a declared overlap domain. Exact fixed-point correspondence then follows from conjugacy; approximate commutation gives only a residual and an error budget.

This paper is intentionally interpretive. The technical conditional conjugacy and unstable-subspace results belong to the MTT–asymptotic-safety paper. The truncation residual bounds belong to its companion truncation paper. The worldsheet obligations belong to the q79 heterotic program.

2 Corner one: selected coherent geometry

The upper corner is not an unspecified “coherent spine.” Its strongest current realization is the selected q79 branch. At the finite tier the corpus contains:

- a finite root-stack and shared-line construction;
- a $1 + 2 + 3$ rank filtration;
- exact finite projectors and a normalized finite Hessian;
- a rank-two coherent kernel and rank-four penalized complement;
- finite Standard-Model carrier and profile-level compatibility results.

These are real algebraic achievements. They do not yet provide the selected physical nonzero-Chern HYM bundles, continuum Hessian blocks, upper action, or four-dimensional quantum theory. The continuum naturality theorem must preserve connections, covariant derivatives, sectors, and Hessians, not just vector-space dimensions.

The phrase “upper” therefore means prior to the effective encodings, not already proven more fundamental physics.

3 Corner two: the q79 string branch

Fu–Yau geometry gives a mathematically established route to non-Kähler heterotic compactifications satisfying the Hull–Strominger system under specific bundle and anomaly conditions [1, 2]. This makes the q79 Fu–Yau branch the strongest current MTT compactification candidate.

The present MTT worldsheet contract is not complete. Its current ledger has:

$$5 \text{ available rows}, \quad 2 \text{ partial rows}, \quad 5 \text{ open rows}.$$

The missing work includes the selected physical visible-hidden bundle, common stability chamber and HYM connections, full anomaly/Bianchi compatibility at the selected endpoint, and the infrared SCFT/GSO data.

Consequently, “string corner” means a selected geometric candidate with a partial contract. It does not mean that perturbative string theory, the full worldsheet CFT, and four-dimensional MTT have already been proved equivalent.

4 Corner three: asymptotic safety

The FRG describes the scale dependence of an effective average action Γ_k [3, 4]. Its asymptotic-safety program searches for a dimensionless ultraviolet fixed functional with finitely many relevant directions.

The revised MTT asymptotic-safety work now provides four conditional tools:

1. an integrable-flow theorem for existence of a Banach-space endpoint;
2. an exact-conjugacy theorem for transporting fixed points and spectra;
3. a truncation residual theorem with an a posteriori error bound;
4. a quasi-compactness theorem for a finite-dimensional unstable subspace.

These tools do not yet supply the physical input to which they would be applied. The open data are:

- a selected chart from q79/coherent variables to dimensionless FRG couplings;
- a physical regulator, gauge fixing, ghost sector, and action norm;
- a four-dimensional shell-integrability estimate;
- a certified truncation defect and essential spectral-radius bound.

The third corner is therefore a diagnostic language and a possible target encoding, not an independently closed UV completion.

5 What a valid triple comparison looks like

Let $\mathcal{U}$ denote an overlap domain in the selected upper theory, $\mathcal{W}$ a worldsheet description space, and $\mathcal{F}$ an FRG action space. A comparison requires maps

$$\mathcal{C}_{\text{ws}} : \mathcal{U} \rightarrow \mathcal{W}, \quad \mathcal{C}_{\text{FRG}} : \mathcal{U} \rightarrow \mathcal{F}.$$

Each map must carry more than configurations. It must specify:

- domains and admissible states;
- parameters and normalization conventions;
- symmetries, constraints, and quotient operations;
- scale evolution or fixed-point maps;
- observables used in the comparison;
- exactness or an explicit defect norm.

Suppose $T_{\mathcal{U}}$, $T_{\mathcal{W}}$, and $T_{\mathcal{F}}$ are the relevant step maps. The exact target is

$$\mathcal{C}_{\text{ws}} T_{\mathcal{U}} = T_{\mathcal{W}} \mathcal{C}_{\text{ws}}, \quad \mathcal{C}_{\text{FRG}} T_{\mathcal{U}} = T_{\mathcal{F}} \mathcal{C}_{\text{FRG}}.$$

Under injective regular charts, upper fixed points then map to fixed points and the restricted linearizations are similar. This conclusion is imported from the technical conjugacy theorem.

If instead

$$\|\mathcal{C}_{\text{FRG}} T_{\mathcal{U}} - T_{\mathcal{F}} \mathcal{C}_{\text{FRG}}\| \leq \varepsilon_{\mathcal{F}},$$

the image has a fixed-point residual of order $\varepsilon_{\mathcal{F}}$. A nearby exact fixed point follows only when a separate validation theorem applies. The same rule holds in the worldsheet corner.

6 What can actually be compared

Quantity	Meaningful comparison	Required certificate
Fixed point	Same selected upper state in two charts	Intertwining step maps and domain overlap.
Critical exponents	Spectra of conjugate linearizations	Differentiable chart, isolated spectrum, and error bound.
Relevant directions	Finite unstable Riesz subspace	Essential spectral-radius control.
Anomaly conditions	Worldsheet and spacetime consistency	Selected bundles, connections, and Bianchi identity.
Four-dimensional couplings	Reduced observables in one convention	Reduction, matching, RG, and normalization map.
UV damping	High-mode behavior of a physical operator	Same-source spacetime operator, not an internal filter alone.

Equal numerical values without these certificates can still be suggestive, but they do not establish that the calculations describe the same object.

7 How disagreement would help

The three-corner program is valuable even when the corners disagree. Suppose a selected upper candidate passes the finite q79 constraints but cannot satisfy the worldsheet anomaly conditions. Then either the candidate is not a physical heterotic background or the proposed worldsheet chart is wrong. If the same candidate yields an FRG flow with no admissible fixed point under a proved chart, then the asymptotic-safety interpretation fails for that branch.

This makes the diagram a falsification tool. It does not guarantee that all successful theories are secretly equivalent. It asks whether one selected object survives several independently meaningful representations.

8 Relation to the shared circle and fixed points

The common q79 line can provide shared phase and holonomy data across finite operators. It does not by itself identify a worldsheet renormalization scale, an FRG scale, or Lorentzian time. Those identifications require connection-preserving maps.

Likewise, fixed points appear in all three corners for different reasons:

Upper fixed point

stability or closure under the selected MTT map;

Worldsheet fixed point

vanishing beta functions and conformal consistency;

FRG fixed point

scale invariance of a dimensionless effective action.

The shared word does not establish equality. Conjugacy of the maps does.

9 Current status

Corner or bridge	Status	Current boundary
Finite q79 upper carrier	Exact at finite tier	Physical continuum HYM/action transfer open.
q79 heterotic worldsheet	Partial, 5/12 available	Visible-hidden endpoint and IR worldsheet data open.
FRG analytic toolkit	Exact conditional	Physical chart and estimates open.
Upper-to-worldsheet map	Partial/open	Full same-source worldsheet contract absent.
Upper-to-FRG map	Open	No selected coupling chart or conjugacy certificate.
Triple diagnostic	Proposed	Usable once both maps have certified overlap domains.
Triple equivalence of UV completions	Not established	Neither pairwise bridge is complete.

10 Discussion

The revised picture is less dramatic and more informative. The q79 branch anchors the geometric side in established heterotic mathematics, while the FRG corner supplies a language

for scale dependence and relevant directions. MTT contributes the possibility that both are representations of one selected upper construction. The research value lies in constructing and testing the arrows.

A particularly useful next target is a small overlap packet rather than a claim of total equivalence. Choose one selected q79 fluctuation sector, derive its reduced quadratic operator, identify a finite set of dimensionless FRG couplings, and compute both the worldsheet and FRG linearized responses with conventions and errors. Agreement would be a nontrivial cross-check. Disagreement would localize the broken bridge.

11 Conclusion

Asymptotic safety can function as a third corner of the MTT quantum-gravity program, but today it is a conditional diagnostic. The finite q79 geometry, partial heterotic contract, and conditional FRG mathematics are mutually relevant without yet being equivalent.

The correct forward statement is therefore:

one selected upper object $\longrightarrow$ several certified encodings $\longrightarrow$ cross-checked observables.

The next advance must construct one of those certified arrows. Redrawing the triangle more strongly would not move the frontier.

Computational Evidence and Reproducibility

The comparison between the asymptotic-safety and string routes is a conditional structural diagnostic. No numerical Standard Model packet proves that comparison. The mapped strict-upgrade ledger is included only to mark the unresolved no-knob boundary inherited by later ultraviolet claims.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>

Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a

Manifest: release/result_manifest.json

Manifest SHA-256:

fb399689 60b00584 631dbf53 1a708e18

ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).

Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] A. Strominger, “Superstrings with torsion,” *Nuclear Physics B* **274** (1986), 253–284.
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- [4] M. Reuter, “Nonperturbative evolution equation for quantum gravity,” *Physical Review D* **57** (1998), 971–985.
- [5] P. Nero, *Modal Triplet Theory: Foundations*, current revised MTT paper corpus, 2026.
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- [7] P. Nero, *Asymptotic-Safety Truncations as Conditional Shadows: Fixed-Point Residuals, Error Bounds, and Scheme Dependence*, current revised MTT paper corpus, 2026.